

A Mathematical Model for Improving Sleep Quality Using Recovery  
Rate Dynamics Based on Exponential Decay.

A Term Paper

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By

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## 1. INTRODUCTION

Sleep is an important psychological process which is essential for human health, cognitive health and overall wellbeing. In today's society, sleep disorders and inadequate sleep have become major public health issues, impacting millions of people across the globe. With the introduction of wearable technology, the monitoring of sleep has turned into a revolution, as it is now possible to continuously track sleep patterns using devices, including smartwatches, Fitbit, Oura rings, and WHOOP bands. These devices collect vast amounts of data including sleep duration, sleep stages, heart rate variability, and other physiological parameters.

Although the use of wearable sleep trackers has become very common, there is a serious gap in the readability and scientific basis of the measures that these devices offer. The majority of the commercial sleep tracking apps are based on internal algorithms that produce sleep scores based on statistical tools and heuristics. Nonetheless, such scores tend to be unclear and not mathematically rigorous such that users and scholars are likely to find it challenging to comprehend the mechanisms of sleep recovery. This is because the lack of mathematical models restricts our capacity to make evidence-based assumptions on the dynamics of recovery and also customize individualized sleep advice.

Sleep recovery is the slow recuperation of physical and intellectual functions and abilities that have been emptied through out the day. This is not a quick process but can take place gradually during the sleep period. The mathematical framework of sleep recovery needs to be able to reflect the ongoing nature of physiological restoration. In the recovery process, there are characteristics indicating that it can be modelled on the basis of the dynamical systems theory especially on exponential decay processes that are normally witnessed in biological systems.

This gap is bridged by the current research that creates a clear mathematically sound model of sleep recovery in terms of recovery rate dynamics. This research instead of viewing sleep recovery as an isolated event or producing opaque numerical measures, conceptualizes recovery

as a time-dependent process that is defined by a set of differential equations. The model models sleep recovery as a recovery deficit variable that declines exponentially with time, which is purely based on the intuitive fact that recovery during sleep is greatest at the initiating stages of sleep and cumbersome as the system approaches equilibrium.

The mathematics that is used in this research is based on the first-order linear ordinary differential equations, which is a proven instrument in applied mathematics to describe decay processes in nature. A number of benefits come with the model such as analytical tractability, computational efficiency, and interpretability by posing sleep recovery as an initial value problem having exponential decay dynamics. The model has a parameter of recovery rate that can be individualized, depending on each person and sleep characteristics, meaning the framework can allow the inter-individual variability without a fundamental change of the model structure.

The study fills this gap between theory and practice in sleep science. The model is made computationally practical; that is, it can be implemented through ordinary numerical tools and computing systems like Python. The exponential decay formulation reflects the major qualitative issues such as initial recovery that is rapid, slowing down as the number of naps increases, and approaching full recovery asymptotically. These attributes are in line with empirical findings of sleep studies as well as physiological research.

This study is not limited to formulating a mathematical model. It provides a base to further extensions such as stage-specific recovery dynamics, stochastic modelling to capture variability, and the combination with wearable real-world data to estimate the parameters and validate them. This work can advance the more comprehensive objective of creating evidence-based, patient-centred strategies of sleeping health optimization by offering a clear mathematical explanation of how sleep recovery works.

## 2. OBJECTIVES

The primary aim is to develop and analyse a mathematical model that describes sleep recovery dynamics using exponential decay principles. The specific objectives of this study are as follows:

### Objective 1: Critical Analysis of Current Sleep Tracking Methodologies

To prove that the current wearable sleep tracking devices, despite being able to provide a rather detailed amount of data, do not provide transparent mathematical methods of explaining the metrics of recovery and sleep scores. This goal includes discussing the drawbacks of proprietary algorithms and a necessity in the open and interpretable mathematical models.

### Objective 2: Deal with the problem of Overdependence on Heuristics

To critically assess the present excessive reliance on statistical and heuristics techniques in the creation of sleep scores, and to suggest a new model, based on the principles of mathematical modelling. This involves the detection of certain deficiencies of black-box algorithms regarding scientific rigor and user interpretability.

### Objective 3: On-going Process Modelling.

To specify the model of sleep recovery as a continuous, time dependent on process, instead of a discrete event or a fixed score. The given objective focuses on the significance of recording the temporal dynamics of physiological recovery during the whole sleep period.

### Objective 4: Recovery rate Dynamics Framework.

To describe sleep-recovery mechanisms in terms of simple, understandable rate dynamics of a recovery process, in the form of a set of differential equations. It includes the development of the recovery process as rate equations which discuss the change in the recovery deficit with time.

### Objective 5: Understandable Mathematical Construct.

To represent a mathematically rigorous framework of understanding sleep recovery accessible to both researchers and also available to educated users and healthcare practitioners. The framework ought to be of sufficient mathematical sophistication and practical interpretability.

Objective 6: Analytical and Numerical Solution Methods.

To obtain analytical solutions to the model proposed differential equations and construct numerical approximation schemes, which allow implementing and simulating the dynamics of recovery process to be computed and simulated.

Objective 7: Model Validation and Interpretation

To confirm the qualitative behaviour of the model with simulation and show that it agrees with the well-known physiological principles of sleep recovery, such as a rapid initial recovery and a slowing down process.

Goal 8: Future Extensions Foundation.

To provide a basic model that can be generalized to achieve more complexity like sleep phase specific recovery rates, stochastic variability, and be able to combine with empirical data provided by wearable devices.

### **3. REVIEW OF LITERATURE**

Sleep science, wearable technology, and math modelling have become the subject of growing interest in recent years. This part is a review of relevant literature in various fields to put the current study in perspective.

#### **3.1. Sleep Physiology and Recovery Mechanisms**

Sleep is a complicated physiological state associated with the decreased consciousness, low level of sensory activity, and specific electrophysiological patterns. The two-process system of sleep regulation proposed by Borbely (1982) is one of the most powerful theoretical frameworks of sleep studies. This theory suggests two independent processes that control sleep: a homeostatic process (Process S) that rises when people are awake and falls when they are asleep, and a circadian one (Process C) that oscillates with the period of about 24 hours. The homeostatic aspect of this model models the sleep need or requirement as accumulating when awake and dissipating when asleep, which gives a theoretical basis of modelling the dynamics of recovery. Achermann and Borbely (2003) have also discussed further the mathematical modelling of the two-process model where the exponential functions would be useful in modelling the homeostatic sleep drive. Their work confirmed that sleep pressure changes exponentially in the wake state and exponentially in the sleep state, and the time constants characterize differences between sleep regulation of individuals. This mathematical analysis gave significant precedence to the use of exponential decay models in the recovery of sleep.

Dijk and Beersma (1989) examined the dynamics of slow-wave activity, in sleep and found that the electroencephalographic assessments of the degree of sleep exhibit exponential attenuation curves throughout the sleep duration. Their results bear out the idea that recovery mechanisms that can be experienced in the course of sleep have mathematical predictable patterns and recovery is greatest at the initial stages of sleep.

### 3.2. Wearable Sleep Tracking Technology.

Growth of consumer wearable devices has made sleep monitoring a laboratory experiment to an available daily activity. Depner et al. (2020) provided a general overview of wearable sleep technology and analysed the accuracy and reliability of different commercial devices. Their review showed that although modern wearables can achieve a reasonable accuracy in measuring the duration of sleep, as well as the sleep wake transitions, the proprietary algorithms employed in creating sleep quality scores are not scientifically validated and transparent in most cases.

A systematic review and meta-analysis of wearable sleep trackers versus polysomnography as the gold standard of sleep assessment were conducted (Haghighat et al., 2019). They concluded that consumer devices exhibit moderate levels of agreement of total sleep time but have shortcomings in the ability to distinguish sleep stages accurately. Notably, they emphasized the non-standardization of the way the devices compute the sleep scores and recovery metrics, which contributes to the necessity of the mathematical models that should be transparent.

The results of sleep tracking devices in identifying the sleep stages such as rapid eye movement (REM) sleep, deep sleep, and light sleep was directly studied by De Zambotti et al. (2019).

Although they agreed that sensor technology and algorithms sophistication is improving, they highlighted that the mathematical basis that underpins proprietary sleep scores are mostly obscure to users as well as researchers. This is because wearable data is not useful in scientific research and evidence-based sleep optimization due to this transparency.

### 3.3. Mathematical Modelling of sleep.

The history of mathematical modelling of sleep has a strong background, but most models have concentrated on circadian rhythms and sleep-wake regulation as opposed to dynamics of recovery. The model of the physiological dynamics of sleep-wake interaction described by Phillips and Robinson (2007) in terms of the interaction of sleep-promoting and wake-promoting neuronal populations is a mathematical model, presented in form of coupled differential equations. Their contribution proved the ability of dynamical systems methods to describe the complicated dynamics of sleep control.

Skeldon et al. (2017) used mathematical modelling to comprehend the effects of light exposure on sleep patterns, in which they attempted to model the circadian pacemaker and how it is entrained by the effects of environmental light through the use of differential equations. Their

study has emphasized the role of individual parameters in models of sleep, because the variability of circadian period and light sensitivity in individuals has a great effect on the predictions of the model.

Postnova et al. (2018) conducted a review of mathematical models of sleep regulation and circadian rhythms and offered a broad summary of the modelling methods of phenomenological models to detailed neurophysiological simulations. They stressed that model complexity should be dictated by the nature of the research questions and the available data, and recommended the use of parsimony when the simpler models were sufficient to model the key dynamics.

### 3.4. The Exponential Decay Models of the Biological Systems.

The exponential models of decay are so common in biological and physiological modelling because of their mathematical manageability and common agreement with empirical data.

Batschelet (1979) gave mathematical foundations to exponential processes in biology, showing that a great deal of physiological variables have first-order kinetics of exponential decay or growth.

Banister et al. (1975) created a mathematical model of athletic training and recovery based on exponential decay functions to model the decay of fitness as well as fatigue. Their theory of impulse-response model conceptualized training in the form of discrete impulses which elicit positive effects (fitness) and negative effects (fatigue), and both have an exponential-decaying effect with time. The modelling framework has been widely used in the sports science discipline and offers conceptual analogies to sleep recovery modelling.

Busso (2003) generalized the fitness-fatigue modelling by adding the concept of different decay rates of variables and proved the significance of parameterization of specific variables in order to predict the performance dynamics accurately. The similarities between athletic recovery and sleep recovery hint at the possibility of the similar mathematical techniques being productive in the field of sleep.

### 3.5. Gap in Current Literature

Although the literature reviewed has shown significant advancement in the field of sleep physiology, wearable tech, and mathematical modelling, a gap in the literature still exists. Current mathematical models of sleep are mostly concerned with the regulatory mechanisms that govern

the transitions between sleep and wakefulness and circadian rhythms and not with the processes of rest that go on when one sleeps. This gap is closed by the current study which formulates an explicit, available mathematical model which is used to describe the dynamics of sleep recovery that apply the concepts of exponential decay.

## **4. RATIONALE OF THE STUDY**

### **4.1. Research Problem**

Although the wearable sleep tracking devices have become a common thing, and already billions of data points have been generated concerning human sleep patterns, there is a fundamental disconnection between the data gathered and our capacity to interpret it in any meaningful way. The existing wearables show customers sleep scores and recovery measurements, which is determined by hidden, proprietary algorithms. Such metrics can affect significant resolutions regarding health practices, work timetables, and doctor visits, but the consumers and even most healthcare gurus do not investigate the mathematical basis of such recommendations. The research question to be pursued in this work can be formulated in the following way: it has no publicly available, mathematically understandable model as per the mechanism of sleep recovery over time, which can be adopted into practice using wearable sleep data. This lack restricts science and practice of sleep recovery principles.

### **4.2. Why This Topic Was Selected**

The choice of this research topic was caused by a number of considerations which are interconnected:

**Scientific Gap:** The lack of a clear linkage between the data abundance and the interpretive paradigms is a central issue of the contemporary sleep science. A solution to this gap is provided by the modelling of the recovery dynamics as they result in explicit, testable hypotheses.

**Practical Relevance:** Millions of people utilize wearable sleep trackers in their everyday lives making decisions on the basis of metrics offered by these devices. An explicit mathematical

model may lead to better interpretability and usefulness of wearable sleep data, which could promote a better level of sleep health at scale.

**Mathematical Interest:** Sleep recovery is an area of study which has properties that are amenable to mathematical modelling such as continuous time evolution, rate dependent dynamics, and can be parameterized by individuals. The use of differential equations in the field is an intellectual challenge of applied mathematics and physiological science.

**Basic Development:** This study provides a baseline model that is simple and easy to interpret to give the basis on which more complex models can be derived. The exponential decay model can be logically expanded with other complexities as are justified by empirical evidence and research questions.

#### 4.3. Operational Definitions

**Recovery Deficit  $R(t)$ :** A continuous variable that is the amplitude of physiological and cognitive recoveries required at time  $t$  with  $t = 0$  being the time of sleep onset. When recovery takes place, the recovery deficit approaches a value of  $R_0$ .

**Recovery Rate Parameter  $k$ :** This is a positive constant that defines the speed of recovering. The greater  $k$  is, the faster the recovery, the lower is  $k$ , the slower is the recovery dynamics. This parameter can differ among different people depending on physiological attributes.

**Exponential Decay:** A mathematical phenomenon where a quantity varies at a rate proportional to difference between the current quantity, leading to rapid change at the beginning and a slowing down towards an end value that is asymptotic to the original.

**Sleep Recovery:** This is the gradual restoration of physiological and cognitive functioning that has been drained during the waking state, and this is done during the sleep period through the restorative processes in the body.

#### 4.4. Limitations of the Study

There are a number of notable weaknesses associated with this research that must be taken into account during the interpretation of the results:

**Model Simplification:** The exponential decay model is an abstracted simplification of complicated physiological processes. Various systems interact to bring about real sleep recovery such as metabolic restoration, hormonal regulation, synaptic homeostasis, and memory consolidation. All this cannot be summarized by the single-equation model.

**Lack of Empirical Support:** The study constructs and then tests a mathematical model of theoretical importance, but does not provide empirical testing in comparison to empirical data concerning actual wearable sleep or polysomnographic performance. This initial theoretical work did not have access to datasets that are needed to estimate parameters and test models as well.

**Uniform Recovery Assumption:** The model also assumes that the recovery rate is constant during sleep, but in the real sleep, there are stage specific recovery properties. Deep, REM, and light sleep probably play a varied role in the different facets of restoration.

**Individual Variability:** It is also true that the model involves a personalized parameter,  $k$ , although the other sources of inter-individual variability are not explicitly taken into consideration by the framework (age, health condition, sleeping disorders, or circadian phase).

**External Factors:** The model fails to integrate environment and behavioural variables which can affect the recovery of sleep, including room temperature, noise, intake of caffeine, or stress levels.

#### 4.5. Research Hypotheses

The hypotheses in this research will be based on the theoretical framework and the literature reviewed:

Hypothesis 1: Sleep recovery is meaningfully modelled as a continuous time-varying process, which is exponential decay in nature.

Hypothesis 2: The first-order linear ordinary differential equation would offer a mathematically solvable model with the necessary qualitative properties of the sleep recovery process such as rapid initial recovery and slowing down.

Hypothesis 3: The recovery rate parameter  $k$  is personal and can be used to customization sleep recovery models.

Hypothesis 4: The recovery trajectories can be computed and simulated by the numerical approximation of the continuous recovery model through the use of finite difference methods.

## 5. METHODOLOGY

### 5.1 Research Design

The paper relies on the theoretical mathematical modelling method to research the dynamics of sleep recovery. The study plan adheres to the systematic flow of the model formulation to the analytical solution, numerical implementation, and qualitative validation. The applied mathematics, the theory of dynamical systems, and fundamental principles of computational science are combined with the application of the methodology which tends to form a complete framework of how to understand the concept of sleep recovery.

### 5.2 Data Sources and Variables

The model is to be worked with sleep data which can be received in different sources:

**Wearable Devices:** Commercial sleep tracking watches such as Smart watches (apple watch, Samsung Galaxy Watch), specialized sleep trackers (Fitbit, Oura ring, WHOOP band, etc.), and other consumer-based wearable that capture sleep duration, sleep stages, and physiological indicators.

**Relevant Variables:**

- **Total Sleep Duration:** The summative time of being asleep in a sleep period.
- **Sleep Stages:** The sleep can be divided into three stages: REM (rapid eye movement), deep (slow-wave sleep), and light sleep stages.
- **Sleep Efficiency:** The proportion of the time spent asleep to the time in bed.
- **Heart Rate and Heart Rate Variability:** Physiological data that can possibly be related to recovery strength.
- **Sleep Onset Time, Wake Time:** Temporal indicators of the sleep interval.

**Model Development Data:** Simulated data and estimated values of parameters were used to develop and test the mathematical model. This method enabled the investigation of the model behaviour in a variety of conditions without having to gain access to proprietary datasets.

### 5.3 Formulation of mathematical model

The mathematical model is assembled in the systematic manner as follows:

#### Step 1: Select Variables of the State

The recovery deficit variable  $R(t)$  is used to represent the recovery process, where  $t \geq 0$  is time since the onset of sleep. The use of a deficit variable (preferred to use the recovery score) as a measure of remaining recovery required has a number of benefits. The deficit variable naturally tends to decrease with an initial positive value toward zero which gives intuitive relation to the idea of recovery being achieved with time. This formulation does not also have artificial upper limits as a recovery score variable would have.

#### Step 2: Continuous Time Assumption

Recovery sleep is said to be constantly changing not in jumps. The nature of physiological mechanisms that are involved in sleep recovery is continuous, which explains this assumption. Continuous time formulation enables the use of differential equations as the main tool of modelling.

#### Step 3: Rate-Based Modelling Principle

The model aims at defining the rate of recovery instead of trying to define the state of recovery at each point. Such a rate-based model is similar to the way many biological processes are described in nature, and it offers mechanistic understanding of the recovery processes.

#### Step 4: Proportional Decay Assumption

The assumption is that the recovery is proportional to the current recovery deficit, mathematically:

$$\frac{dR(t)}{dt} \propto -R(t)$$

The negative sign here indicates that the recovery deficit decreases over time. This means that when more recovery is needed (higher deficit), recovery processes operate more intensely, whereas as the recovery approaches completion (lower deficit), the rate of further recovery diminishes.

#### Step 5: Parameter Introduction

A positive constant  $k > 0$  is introduced as the proportionality constant, representing the recovery rate parameter.

This gives the governing differential equation:

$$\frac{dR(t)}{dt} = -kR(t)$$

#### Step 6: Initial Value Problem Formulation

To uniquely determine the solution, an <sup>initial</sup> condition is

specified:  $R(0) = R_0$

where  $R_0$  represents the initial fatigue level or recovery deficit at sleep onset. Together, the differential equation and initial condition constitute a first-order linear initial value problem.

#### 5.4 Analytical Solution Method

The initial value problem is solved analytically using standard techniques from differential equations theory:

Starting from, variables are separated:

$$\frac{dR(t)}{dt} = -kR$$

$$\frac{dR(t)}{R} = -k dt$$

Integrating both sides:

$$\int \frac{dR(t)}{R} = \int -k dt$$

$\ln|R| = -kt + C$ , where  $C$  is an integration constant.

Taking exponential of both sides:

$$R(t) = e^{(-kt+C)} = e^C \cdot e^{(-kt)}$$

Let  $A = e^C$ ,  $A$  is an arbitrary constant:

$$R(t) = A \cdot e^{(-kt)}$$

Now, applying the initial condition:

Using  $R(0) = R_0$

$$R_0 = A \cdot e^0 = A$$

Therefore,  $A = R_0$ , gives the complete solution:

$$R(t) = R_0 \cdot e^{(-kt)}$$

This analytical solution describes the recovery deficit as exponentially decaying from its initial value  $R_0$  with decay rate  $k$ .

### 5.5 Numerical Approximation Scheme

Although the analytical solution gives the precise values, the numerical methods are significant due to various reasons: they can be implemented computationally, they can be extended to model of more complex characteristics in which the analytical solution may not be available, and they have a parameter estimation framework based on discrete data. Euler method is used as a simple numerical approximation scheme.

Time Discretization:

Continuous time is discretized using a fixed time step  $\Delta t$ , with discrete time points:

$$t_n = n \cdot \Delta t, n = 0, 1, 2, \dots$$

The recovery deficit at time  $t_n$  is denoted  $R_n \approx R(t_n)$ .

Finite Difference Approximation:

The time derivative is approximated using a forward finite difference:

$$\left. \frac{dR(t)}{dt} \right|_{t=t_n} \approx \frac{(R_{n+1} - R_n)}{\Delta t}$$

Substituting the finite difference approximation into the differential equation:

$$\frac{(R_{n+1} - R_n)}{\Delta t} = -k \cdot R_n$$

$$\text{Solving for } R_{n+1} = R_n - k \cdot R_n \cdot \Delta t = R_n(1 - k \cdot \Delta t)$$

### 5.6 Simulation Parameters

For model simulation and qualitative validation, these values were selected to illustrate model behaviour across a range of recovery rates while maintaining numerical stability of the Euler method:

- Initial Recovery Deficit:  $R_0 = 100$  (arbitrary units)

- Time Step:  $\Delta t = 0.1$  hours (6 minutes)
- Sleep Duration: 8 hours
- Recovery Rate Values:  $k \in \{0.2, 0.5, 1.0\}$  per hour

### 5.8 Model Validation Approach

The qualitative validation of the model is aimed at ensuring that the simulated recovery paths have the characteristics that are theoretically expected:

- Monotonic Decrease: The recovery deficit is to be decreasing monotonically.
- Quick Initial Recovery: The greatest change in recovery should occur early in sleep.
- Asymptotic Behaviour: Recovery deficit must tend towards zero with time.
- Sensitivity of parameters: An increase in  $k$  values is expected to result in accelerated recovery.

## 6. RESULTS AND DISCUSSION

### 6.1 Analytical Solution Properties

The analytical solution  $R(t) = R_0 \cdot e^{(-kt)}$  shows many important mathematical properties that align with physiological expectations for sleep recovery:

**Exponential Decay:** The solution shows pure exponential decay, a well-characterized mathematical behaviour. At  $t = 0$ ,  $R(0) = R_0$  as required by the initial condition. As time progresses, the exponential term  $e^{(-kt)}$  decreases monotonically, driving the recovery deficit toward zero.

**Asymptotic Behaviour:** Due to the fact that, as  $t \rightarrow \infty$ ,  $e^{(-kt)} \rightarrow 0$ , we can conclude that no limit exists to  $R(t)$ , which tends to 0. This mathematical characteristic is related to the physiological meaning of full rest within an adequate period of time. The asymptotic approach shows that recovery is carried out in such a way that recovery is never actually completed but rather it tends toward completeness.

**Time Constant:** The characteristic time constant of the system is  $T = 1/k$ . This quantity represents the time required for the recovery deficit to decrease to about 37% (specifically,  $1/e \approx 0.368$ ) of its initial value. For example, with  $k = 0.5$  per hour, the time constant is  $T = 2$  hours, meaning that after 2 hours of sleep, roughly 63% of recovery has occurred.

**Half-Life:** The time required for the recovery deficit to decrease to half its initial value is  $t_{1/2} = \ln(2)/k \approx 0.693/k$ . This provides an intuitive measure of recovery speed. For  $k = 0.5$  per hour, the half-life is about 1.4 hours.

### 6.2 Numerical Simulation Results

The simulations are conducted for three representative recovery rate values:  $k = 0.2$ ,  $k = 0.5$ , and  $k = 1.0$  per hour, all with initial deficit  $R_0 = 100$  and sleep duration of 8 hours.

Case 1: Slow Recovery ( $k = 0.2$  per hour)

With a low recovery rate of  $k = 0.2$ , the simulation demonstrates relatively gradual recovery over the 8-hour sleep period:

- After 2 hours:  $R(2) \approx 67$  (33% recovery)
- After 4 hours:  $R(4) \approx 45$  (55% recovery)
- After 8 hours:  $R(8) \approx 20$  (80% recovery)

The slow recovery rate tells that significant deficit remains even after a full night's sleep. This pattern might represent individuals with impaired sleep quality or those recovering from severe fatigue or tiredness.

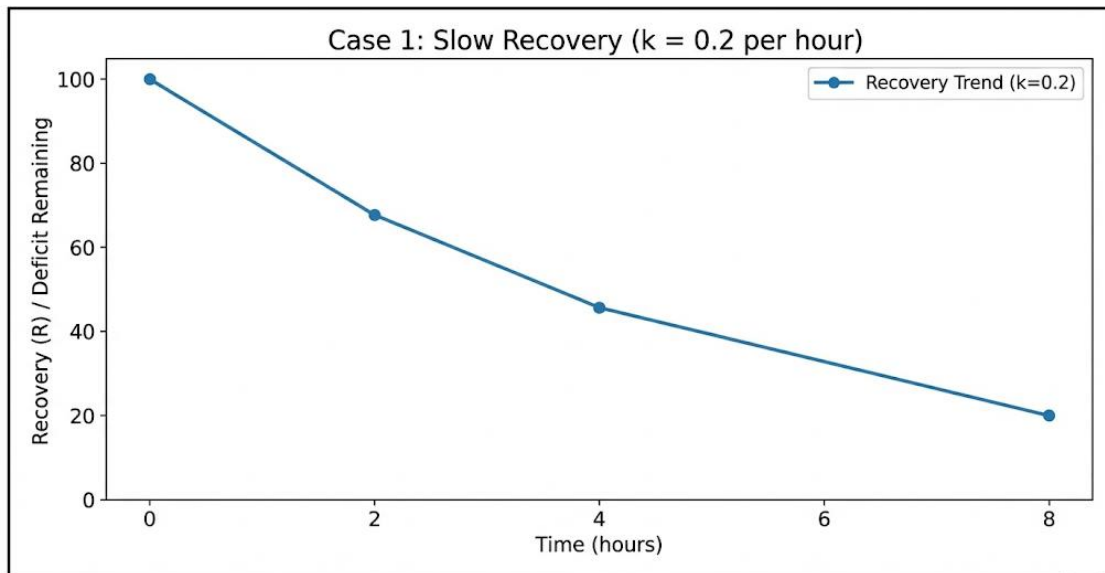


Figure 1: Gradual recovery over an 8-hour period with a low recovery rate.

#### Case 2: Moderate Recovery ( $k = 0.5$ per hour)

With a moderate recovery rate,  $k = 0.5$ , the simulation shows more substantial recovery within the 8-hour period:

- After 2 hours:  $R(2) \approx 37$  (63% recovery)
- After 4 hours:  $R(4) \approx 14$  (86% recovery)
- After 8 hours:  $R(8) \approx 2$  (98% recovery)

This moderate case represents typical healthy sleep, with the majority of recovery occurring in the first half of sleep and near-complete recovery by the morning.

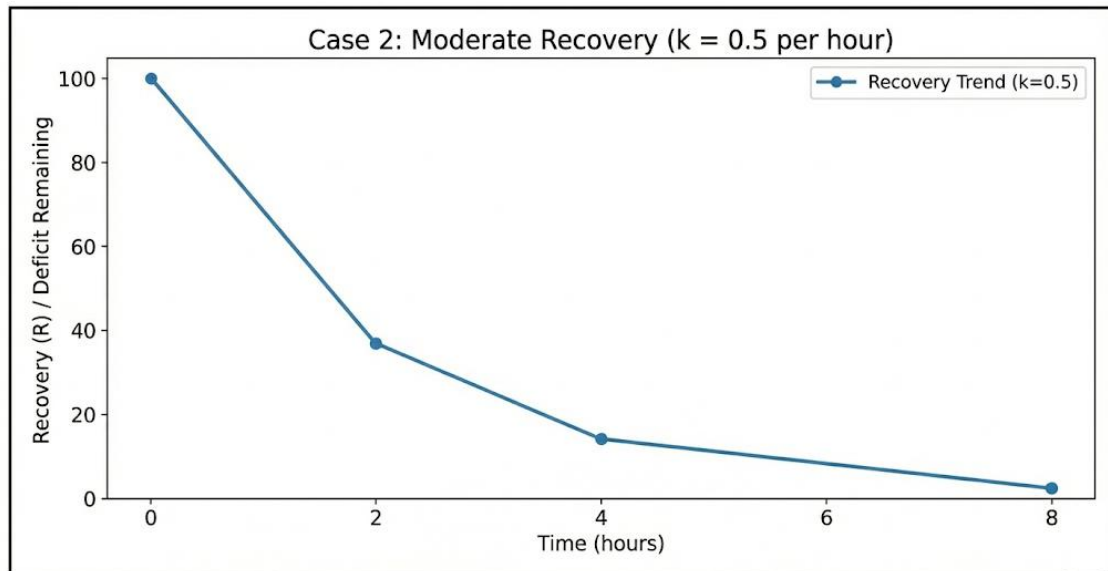


Figure 2: Substantial recovery over an 8-hour period with a moderate recovery rate.

### Case 3: Rapid Recovery (k = 1.0 per hour)

With a high recovery rate of  $k = 1.0$ , the simulation demonstrates very rapid recovery:

- After 2 hours:  $R(2) \approx 14$  (86% recovery)
- After 4 hours:  $R(4) \approx 2$  (98% recovery)
- After 8 hours:  $R(8) \approx 0.03$  (99.97% recovery)

Rapid recovery is usually of individuals with high-quality sleep or young, healthy individuals with efficient recovery processes. The majority of recovery occurs within the first few hours of sleep.

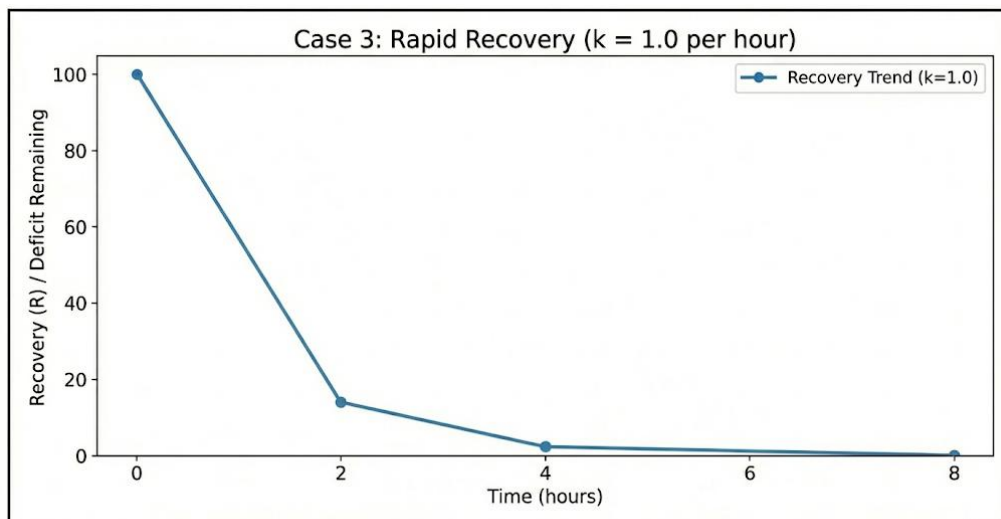


Figure 3: Rapid and near-complete recovery within the first few hours of sleep with a high recovery rate.

### 6.3 Qualitative Validation

#### Monotonic Decrease Confirmed:

In all cases, the recovery deficit  $R(t)$  decreases monotonically from its initial value towards zero.

#### Asymptotic Behaviour Confirmed:

The recovery deficit approaches but never quite reaches zero, which illustrates the mathematical fact of asymptotic convergence. This habit is in line with the fact that even though sleep could offer a great deal of rest, a few percent of fatigue could still be experienced.

#### Rapid Initial Recovery:

The greatest change in recovery deficit value occurs during the early phase of sleep. This clearly aligns with the psychological evidence that deep sleep which predominates early in the night, is particularly restorative.

#### Parameter Sensitivity:

The recovery rate parameter  $k$  produces the expected effect i.e. higher values yield faster recovery and lower values producing slower recovery.

### 6.4 Interpretation and Psychological Relevance

The model aligns with established sleep recovery;

#### Front-Loaded Recovery:

The model's prediction of rapid initial recovery matches with empirical observations about the restorative function of early sleep. The metabolic restoration, release of growth hormone, and downscaling of synapses are linked to slow-wave that is most common during the initial cycles of sleep.

#### Diminishing Returns:

As the recovery process continues, the degree of subsequent healing reduces. The diminishing returns effect also aligns with psychological findings.

## 6.5 Limitations of Current Results

Several limitations of the current results should be acknowledged:

**Absence of Sleep Stage Differentiation:** The model treats recovery as a uniform process throughout sleep, whereas different sleep stages (REM, deep, light) likely contribute differentially to various recovery processes. A more sophisticated model would incorporate stage-specific recovery dynamics.

**Lack of Empirical Calibration:** The simulations use assumed parameter values rather than values estimated from real sleep data. Empirical validation would require fitting the model to observed recovery markers from wearable devices or polysomnography.

**Single Recovery Dimension:** The model represents recovery as a one-dimensional process, whereas sleep likely supports recovery across multiple independent or partially independent dimensions (physical, cognitive, re emotional, etc.). Multi-dimensional extensions would provide greater realism.

**No Stochasticity:** Real sleep exhibits variability and unpredictability. Incorporating stochastic elements would allow the model to represent night-to-night variability and uncertainty in recovery dynamics.

Despite these limitations, the results successfully demonstrate that a simple exponential decay model can capture essential qualitative features of sleep recovery and provide a foundation for more elaborate future models.

## 7. CONCLUSION

This paper successfully discussed about a transparent mathematical model of sleep recovery using exponential decay dynamics. The model, formulated as the differential equation as  $dR/dt = -kR$  with initial condition  $R(0) = R_0$ , provides an analytic framework which captures essential features of sleep psychology.

The analytical solution  $R(t) = R_0 \cdot e^{(-kt)}$  shows that recovery deficit decreases exponentially over time, with recovery rate parameter  $k$  determining the speed of recovery. Three case scenarios showed that the model successfully reproduces rapid initial recovery followed by gradual decrease consistent with sleep architecture where deep sleep is first during night. The importance of early sleep stages is highlighted.

This work addresses a critical gap between data rich wearable technology like whoop, smartwatches, etc. and theory driven mathematical understanding, providing a foundation that is simultaneously simple enough for real world implementation.

## 8. IMPLICATIONS

There are a number of implications of this model. In theory, it contributes to the mathematical basis of sleep science by giving it an explicit and testable framework allowing research to proceed based on hypotheses and relates sleep recovery to general biological concepts. The open algorithm formulation enables peer communication and reproducibility, which is a concern when using black-box proprietary algorithms.

In practice, this framework could be used by the manufacturers of wearable devices to create recoveries that are interpretable. Individually calibrated models might be used and evidence-based recommendations of sleep made by healthcare providers. Real-time implementation is computationally efficient and allows implementation on battery-powered devices.

The parameter of recovery rate may be used as a quantitative biomarker of sleep quality impairment in sleep disorders such as insomnia or sleep apnea. A lower  $k$  value could reflect disorders that become a factor that can be objectively evaluated and followed up on the treatment.

## 9. LIMITATIONS

1. It is a hypothetical development that has not been empirically tested on actual wearable data. Practical application of the recovery trajectories requires parameter estimation based on observed recovery trajectories.
2. The model considers recovery as a homogenous process across a sleep period with no distinction to sleep stages (REM, deep, light) that may have different restorative effects.
3. It is a one-dimensional view of recovery where sleep favors the presence of various processes (physical, cognitive, emotional) that are independent of each other.
4. There is no consideration of external factors such as sleep environment, caffeine, stress and previous history of sleep.
5. The model assumes the uninterrupted sleep without taking into consideration the awakenings and fragmentation during the night.
6. Variability at the individual level (other than the  $k$  parameter, age, sex, health status, genetics) is not explicitly represented.

## 10. SUGGESTIONS FOR FURTHER RESEARCH

The most logical next step for this research is empirical validation. To test the model's practical value, it needs to be fitted to real-world sleep data collected from wearable devices and compared alongside objective measures of recovery, such as cognitive tests, fatigue ratings, and physiological markers. This process is essential for accurately estimating parameters and assessing how well the model predicts actual recovery.

There is also significant room to expand the mathematical framework. Future iterations could introduce more complexity, such as stage-specific recovery rates where the value of  $k$  changes for deep, REM, and light sleep. The model could also be improved by incorporating stochastic elements to handle random variability, or by adding circadian modulation to make  $k$  time-dependent. Additionally, it would be valuable to account for sleep fragmentation (awakenings) and to explore how machine learning could help estimate personalized  $k$  values based on individual characteristics.

Beyond the model itself, longitudinal studies would provide insight into how recovery dynamics change over time with age, health status, and lifestyle interventions. Validating the model across different populations would also ensure it is robust and broadly applicable. From a clinical perspective, trials could determine if the model is effective in predicting treatment responses for patients with sleep disorders.

Finally, future work should directly compare this approach against existing sleep scoring algorithms to highlight its specific advantages in transparency and interpretability. To encourage adoption by other researchers and practitioners, developing and releasing open-source software implementations of the model would be a highly practical contribution.

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